

FROM SCRATCH 1 TO 4

Data Ingestion

We have several loaders in LangChain

Retrieval Augmented Generation: Process of optimizing LLM output

so it will refer to authorized knowledge base outside its training data source

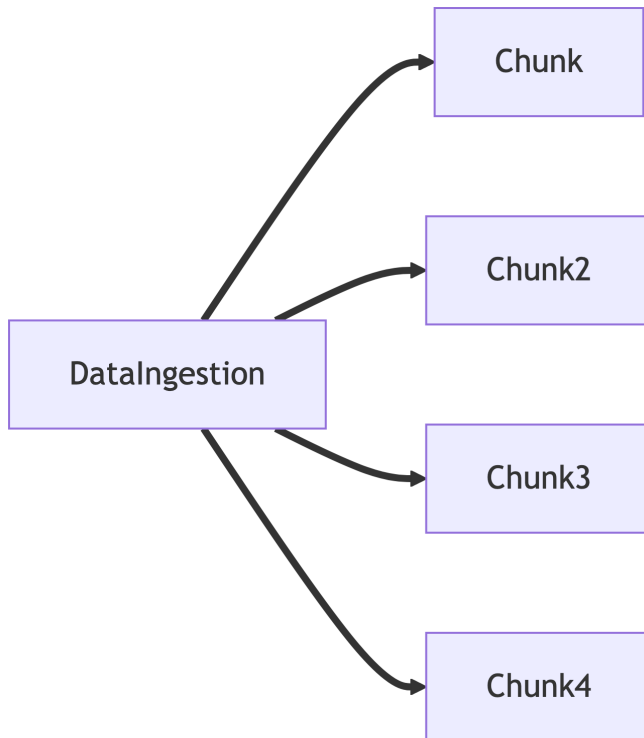
```
from langchain_core.documents import Document
```

getting the query from user and do embedding operation on it and refers out vector
db RETRIEVAL PIPELINE

we have much more type of RAG so like Agentic RAG

Augmentation means we give context to llm and generate the output **Data Ingestion and Data Parsing**

```
#
```



so for every chunk we apply embeddings and finally store it into our final vector db

- **embedding:** convert text to vectors

```
### document structure
from langchain_core.documents import Document
# if we wanna improve our RAG we shall use metadata
doc=Document(
    page_content="this is the main text content I am using to create RAG",
    metadata={
        "source": "example.txt",
        "pages":1,
        "author":"Khrisk Naik",
        "date_created": "2025-01-07",
    }
)

doc
```

```
Document(metadata={'source': 'example.txt', 'pages': 1, 'author': 'Khrisk Naik', 'date_created': '2025-01-07'}, page_content='this is the main text content I am using to create RAG')
```

```
# create a simple txt file
import os

os.makedirs("./data/text_files", exist_ok=True)
# using one dot will create this folder in current pwd
```

```
sample_texts={
    "./data/text_files/python_intro.txt": """Python Programming Introductio

Python is a high level, interpreted programming language known for its simplicity and readability.
Created by Guido van Rossum and first released in 1991, Python has become one of the most popular
Programming languages in the world.

Key Features:
- easy to learn and use
- extensive standard Library
-

python also widely used in web development, data science, artificial intelligence, and automation.

"""
}

for filepath, content in sample_texts.items():
    with open(filepath, 'w', encoding='utf-8') as f:
        f.write(content)

print(" Sample text files is created")
```

Sample text files is created